

Mohammad Nasr

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Robotics PhD student working on embodied and agentic AI and robotics with direct applications in industry. Earlier I worked on nonlinear dynamics and vibration, developing a system for autonomous testing of vibrating systems that makes characterizing nonlinear, real-world dynamical behavior far more tractable. In acoustics, I have worked on binaural hearing, specifically animal auditory systems, leading to potential bioinspired hearing designs. For me, working across disciplines isn't a detour, it's how I think.

EDUCATION

Ph.D., Robotics & Autonomous Systems

Arizona State University — Expected 2029

M.Sc., Mechanical Engineering

Univ. of Nebraska-Lincoln — 2024

Advisor: Keegan J. Moore, Ph.D. — Dynamics & Vibration

B.Sc., Mechanical Engineering

University of Tabriz — 2021

Advisor: Mehran Mahboubkhah, Ph.D.

TECHNICAL TOOLKIT

ROBOTICS

ROS / ROS 2, MoveIt, SLAM, Motion Planning, Manipulation, Sensor Fusion

ML / AI

PyTorch, Transformers, Vision-Language Models, LLMs, Reinforcement Learning, Evaluation Pipelines

SIMULATION

Gazebo, MuJoCo, Isaac Sim, MATLAB/Simulink, Digital Twins, Benchmarking

CONTROL / MATH

Optimization, System Identification, State Estimation, Numerical Methods, Linear Algebra, Statistics

PROGRAMMING

Python, C++, MATLAB, Linux, Git, Data Analysis

RESEARCH OUTPUT

2 journal publications · 2 conference presentations — see p.2

"Smart Automatic Modal Hammer," J. Vib. Eng. Technol. (2024)

PROFESSIONAL EXPERIENCE

Co-founder & Chief Product Designer

May–Sep 2021

VISION Company, Tabriz, Iran

- Designed automatic hand and foot disinfectant devices using SolidWorks, adopted in schools, restaurants, and stores.
- Programmed Arduino Uno as the main control unit for the automation process.
- Generated nearly **30%** revenue through product sales.
- Designed an automatic face mask delivery machine for malls, bus stops, and train stations.

Aerodynamics & Structural Design Lead

Feb–Jun 2019

Robotic Lab, University of Tabriz, Iran

- Designed fuselage, wings, and tail in SolidWorks; analyzed aerodynamic performance in ANSYS.
- Evaluated structural stability from mass distribution and attachment locations.
- Manufactured the first balsa-wood prototype for the national model airplane competition.

Wing Designer

Aug–Sep 2018

Robotic Lab, University of Tabriz, Iran

- Optimized a pre-designed airfoil and designed a tapered wing in SolidWorks for better efficiency.
- Manufactured the wing via balsa wood and 3D printing, used in the wind-tunnel teaching lab.

Aerial Robotic Team Leader

Dec 2017–Aug 2018

Robotic Lab, University of Tabriz, Iran

- Designed the quad-rotor main frame for the Iran Open Robotics competition.
- Led weekly team meetings and secured financial and lab support from research center supervisors.
- Served as drone pilot; worked with electrical circuits and flight controllers.

ACADEMIC PROJECTS

Active Perception for Object Localization

Jan–May 2026

Arizona State University

- Built a TurtleBot4 active-perception system pairing next-best-view selection with a confidence-scoring algorithm to improve pose-estimation reliability.
- Integrated Nav2 global/local planning with an ORB-SLAM3 + EKF + custom A* fallback for mapless navigation.

Acoustic-Based, Bioinspired Robotic Platforms Design

Aug 2024–Dec 2025

Arizona State University

- Analyzed hearing systems of domestic cats, arctic fox, and aye-aye to design bio-inspired robotic hearing methods.
- Co-authored "Guided by Hearing," accepted in the Journal of Material Evaluations.

Assistive Follower Robot (Load-Bearing)

Jan–May 2025

Arizona State University

- Built a TurtleBot 4 follower robot with depth camera, LiDAR, and ROS 2 for hands-free tracking and obstacle avoidance.
- Trained a lightweight ResNet18 model for real-time shoe detection on Raspberry Pi.

Solving a 4x4 Maze with MyCobot Pro 600

Aug–Dec 2024

Arizona State University

- Implemented the A* algorithm for maze solving and robot motion planning, verified via digital-twin simulation.

MORE PROJECTS

Autonomous Testing of Vibrating Systems
Univ. of Nebraska-Lincoln — Aug 2022–May 2024

RL for Robot Navigation (Q-Learning)
Univ. of Nebraska-Lincoln — Jan–May 2024

Autonomous Navigation, F1Tenth Platform
Univ. of Nebraska-Lincoln — Jan–May 2024

Robotic Arm Soccer Player
Univ. of Nebraska-Lincoln — Aug–Dec 2023

Planar Circular Restricted 3-Body Problem
Univ. of Nebraska-Lincoln — Jan–May 2023

TEACHING EXPERIENCE

Teaching Assistant, Robotic Systems 1 Jan–May 2026
Dept. of Manufacturing & Systems, Arizona State University

- Conducted lab sessions using Dobot robotic arms; designed lab exercises and final project assignments.
- Graded student homework assignments and lab reports.

Teaching Assistant, Engineering Dynamics Aug–May 2024
Mechanical & Materials Engineering, Univ. of Nebraska-Lincoln

- Conducted recitation sessions; coached students on course content and problem-solving procedures.
- Corrected and graded student homework problems and exams.

TA, Kinematics & Dynamics of Machinery; Elements of Machine Design Jan–May 2023
Mechanical & Materials Engineering, Univ. of Nebraska-Lincoln

- Addressed student concerns on course materials and solutions; corrected and graded homework and exams.

Teaching Assistant, Kinematics & Dynamics of Machinery Aug–Dec 2022
Mechanical & Materials Engineering, Univ. of Nebraska-Lincoln

- Addressed student concerns on course materials and solutions; corrected and graded homework and exams.

JOURNAL PUBLICATIONS

Nasr, M., Masurkar, N., Meshkizadeh, P., & Dehghan-Niri, E. (2025). Guided by Hearing: Robotic Motion Planning for Precision in Gas Leakage Localization. *Journal of Material Evaluations*. Accepted for publication.

Nasr, M., Singh, A., & Moore, K.J. Smart Automatic Modal Hammer: Predictor–Corrector Approach for Accurate Excitation of Dynamical Systems. *J. Vib. Eng. Technol.* 12 (Suppl 1), 1045–1063 (2024). doi.org/10.1007/s42417-024-01461-6

CONFERENCE PRESENTATIONS

Nasr, M., Masurkar, N., Meshkizadeh, P., & Dehghan-Niri, E. “Gas Leakage Localization Through Bio-Inspired Motion Strategy on Robotic Platforms,” ASNT Research Symposium 2025, Indiana, June 23–27.

Nasr, M., Singh, A., & Moore, K.J. “Smart Automatic Modal Hammer for Studying Nonlinear Dynamical Systems,” International Modal Analysis Conference XLII, Orlando, FL, Jan 29–Feb 1, 2024.

PROFESSIONAL ACTIVITIES

B1, Airframe Mechanics & Aircraft Maintenance Technology Technician Sep 2020–Jul 2021
Civil Aviation Technology College (CATC), Tehran, Iran

- Passed civil aircraft maintenance coursework with EASA modules (European Aviation Safety Agency), covering aerodynamics, structures, gas turbine engines, and propeller systems.

Local Committee Vice President & Organization May–Sep 2017
AIESEC, Tabriz, Iran

- Developed volunteering projects attracting international students to projects in Iran; negotiated financial and accommodation support with local partners.